ACL 2026 · MAIN CONFERENCE

Advancing Speech Tasks Evaluation with Fine-Grained Contrastive Language–Speech Pre-training

Introducing **FCaps** & **CLSP**

ACL 2026 MAIN

Speech Representation | Contrastive Learning | Fine-Grained Captioning | Paralinguistics | Multi-Granular

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Motivation & Problem

EXISTING LIMITATIONS

- Models rely on **coarse discrete labels** — lose temporal dynamics
- Cascaded pipelines: Speech ⟶ tags ⟶ LLM rewrite
  Error propagates; information bottleneck
- Human evaluation: costly, low inter-rater consistency
- LLM-as-judge (GPT-4o, Gemini): expensive, hard to scale

OUR APPROACH

- Speaking style is **multi-scale**: global traits + fine-grained temporal dynamics
- **End-to-end** audio-grounded annotation — no discrete bottleneck
- Multi-granular contrastive training
- Scalable **human-aligned evaluator**

FCaps Dataset & Annotation Pipeline

47k	19M	2	#1
Hours of Speech	Captions	Caption Granularities	Largest Style Dataset

Speech Input audio clip → **Qwen3-Omni** Captioner ×5–20 → **Qwen3-30B** Agentic Verifier → **Verified Captions** multi-positive

Global captions: utterance-level holistic description · **Fine-grained captions**: temporal dynamics, style shifts, prosodic variation

CLSP Model & Two-Stage Training

DUAL-ENCODER ARCHITECTURE

- **Speech encoder**: SPEAR-XLarge (599M) — SOTA SSL
- **Text encoder**: RoBERTa-base (125M) — up to 512 tokens
- MLP projection $\longrightarrow$ shared ℓ_2-normalized embedding space
- Total: **724M parameters**

TWO-STAGE CURRICULUM

- **Stage 1** (1.2M steps): InfoNCE on fine-grained captions
- **Stage 2** (4k steps): Multi-positive InfoNCE with soft targets

Task 1

Global + Fine-grained $\longrightarrow$ cross-granularity generalization

Task 2

Fine-grained × 2 $\longrightarrow$ fine-grained discrimination

Dynamic scheduler: $p_0 = 0.95 \longrightarrow p_{min} = 0.50$ over 10k steps

Exp. ①② — Annotation Quality & Speech–Text Retrieval

① ANNOTATION QUALITY (Gemini 3 Pro, 1k clips)

Pipeline	Correctness	Coverage	Naturalness	Avg.
Cascaded	3.30	3.10	4.15	3.51
End-to-End (Ours)	**4.42**	**4.55**	**4.92**	**4.63**

Largest gain on coverage (+1.45) — directly reflects the information bottleneck in cascaded pipelines.

② SPEECH–TEXT RETRIEVAL (241 clips, mAP@10)

System	Global		Fine-Grained	
	S⟶T	T⟶S	S⟶T	T⟶S
LAION-AI CLAP	1.5	1.9	1.6	1.7
GLAP	3.4	3.9	9.1	4.4
ParaCLAP	3.5	2.3	4.1	3.3
CLSP (Ours)	**58.7**	**54.5**	**77.9**	**77.2**

Exp. ③ — Zero-Shot Paralinguistic Classification

No task-specific training — cosine similarity to natural language prompts. Reported as WA (%) / UA (%).

System	Emotion · IEMOCAP (4-cls)		Emotion · RAVDESS (8-cls)		Emotion · CREMA-D (6-cls)		Gender · RAVDESS		Age · CREMA-D	
	WA	UA	WA	UA	WA	UA	WA	UA	WA	UA
LAION-AI CLAP	32.6	29.1	14.4	13.5	21.0	19.2	58.6	58.6	1.3	1.8
GLAP	32.5	27.0	7.8	13.0	13.6	17.9	72.6	72.6	27.1	32.6
ParaCLAP	46.1	46.5	28.1	30.3	29.8	29.6	99.2	99.2	31.2	32.0
Auden-Voice CLAP	—	—	32.4	—	30.2	—	95.6	—	38.5	—
CLSP (Ours)	**57.2**	**56.1**	**46.8**	**46.0**	**35.1**	**37.2**	**100.0**	**100.0**	**40.6**	**44.5**

Exp. ④ — Speech Style Similarity & Human Alignment

20 expert annotators · Pearson r / Spearman ρ / Kendall τ (all $p < 0.01$)

System	Intrinsic Traits			Situational Traits			Fusion		
	r	ρ	τ	r	ρ	τ	r	ρ	τ
LAION-AI CLAP	0.679	0.664	0.467	0.194	0.184	0.126	0.588	0.597	0.425
GLAP	0.372	0.340	0.234	0.169	0.138	0.102	0.350	0.344	0.241
ParaCLAP	0.663	0.634	0.445	0.323	0.330	0.232	0.626	0.599	0.417
CLSP (Ours)	**0.893**	**0.858**	**0.668**	**0.903**	**0.878**	**0.694**	**0.886**	**0.858**	**0.670**

CLSP tracks absolute quality linearly (r ≈ 0.89–0.90)

Preserves relative ranking (ρ ≈ 0.86–0.88)

Scalable alternative to LLM-as-judge

Conclusion

CONTRIBUTIONS

- **FCaps**: 47k hrs, 19M captions — largest fine-grained speaking style dataset
- End-to-end audio-grounded annotation pipeline with agentic verification
- **CLSP**: dual-encoder, two-stage multi-granular contrastive pre-training (724M)
- Scalable **human-aligned evaluator** — practical alternative to LLM-as-judge

We advocate a shift from **predefined discrete attributes** toward **open-vocabulary, speech-grounded natural language descriptions** for speaking style modeling.

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HuggingFace: yfyeung/FCaps
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github.com/yfyeung/CLSP
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